



Aikaterini Christoforaki

Mechanical Engineer

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in www.linkedin.com/in/katerina-christoforaki 📍 Heraklion Crete, Greece

PROFILE:

Mechanical Engineering undergraduate with a strong passion for motorsport and performance engineering. Hands-on experience in Formula Student powertrain development, vehicle performance analysis, and data analysis. Currently completing a diploma thesis on computational analysis and optimization of an F1 V6 engine block.

TECHNICAL SKILLS:

SIMULATION AND ANALYSIS

ANSYS Thermal - Structural Analysis

Parametric Optimization

CAD & ENGINEERING

SolidWorks · Fusion · AutoCad · N-Top

Parametric Design

DFM & Manufacturing

MOTORSPORTS & DATA TOOLS

MoTeC i2 · Telemetry Analysis · Lap Time Analysis ·

Vehicle Dynamics · Race Engineering Methodology ·

Ricardo Wave · Lotus Engine Simulation

EDUCATION:

HELLENIC MEDITERRANEAN UNIVERSITY 2020 - 2026

Expected Oct/Nov 2026

Mechanical Engineering Diploma

DIPLOMA THESIS:

Advanced Racing Engine Block Design using Computational Modeling.

LANGUAGE:

- English (C2)
- Greek

01 MOTORSPORT EXPERIENCE:



PERFORMANCE ENGINEER | MINAUTOR RACING TEAM

NOV 2025 - TODAY

Formula Student

- Lap time Analysis
- Driver & Team Communication
- Data Analysis
- Vehicle Performance Analysis
- Performance Optimization



POWERTRAIN TEAM LEADER | MINAUTOR RACING TEAM

DEC 2022 - NOV 2025

Formula Student

- Team Management
- Problem Solving
- Critical Decisions
- Drivetrain Analysis
- Communication



CO FOUNDER OF MINAUTOR RACING TEAM

2022 - TODAY

02 ENGINEERING PROJECTS:



ADVANCED RACING ENGINE BLOCK DESIGN USING COMPUTATIONAL MODELING

Diploma Thesis - In Progress

Conducting thermal and structural analysis of a high-performance V6 racing engine block using FEM simulations. Parametric CAD design and optimization focused on minimizing weight, improving heat management, and enhancing engine efficiency and durability.

Tools: N-Top · Ansys · SolidWorks · Matlab



VEHICLE PERFORMANCE & TELEMETRY ANALYSIS

Analysis of vehicle performance using telemetry data. Lap-time analysis and driver feedback to improve overall performance

Tools: MoTeC i2 · Matlab · Excel · Python



DRIVETRAIN SYSTEM ANALYSIS

Formula Student

Developed an Excel-based engineering model to analyse drivetrain performance, gear ratios, and vehicle speed across operating conditions. Optimization of drivetrain setup.

Tools: Ricardo Wave · Excel · Lotus Engine Simulation

03 RELEVANT TRAINING & COURSES:

🎓 RACE ENGINEERING CAP COURSE - MOTORSPORT ENGINEER | TONY ROSS

Completed 2026 on Scholarship | Score: 83%

Race engineering principles, vehicle performance, telemetry data, race-weekend decision making, driver communication and projects.

🎓 VEHICLE DYNAMICS - MOTORSPORT ENGINEER

Completed 2026

Vehicle Dynamics Fundamentals and tyre data analysis, MATLAB-Simulink projects on bicycle model, 4DoF, lateral force, longitudinal dynamics, and 7DoF.

🎓 DATA ENGINEERING IN F3 - MOTORSPORT ENGINEER

Completed 2026

Data analysis, telemetry interpretation and responsibilities of an F3 data engineer.

04 ENGINEERING PORTFOLIO

<https://christoforakiaikaterini.github.io/>